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Analysis of the human auditory nerve

H. Spoendlin and A. Schrott

E.N.T. Department, University of Innsbruck, Austria

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In human temporal bones of patients with normal hearing or sensory neural deafness, the cochlear neurons were quantitatively and qualitatively evaluated at the level of the osseous spiral lamina, the spiral ganglion and the cochlear nerve. We found from 32,000 to 31,000 myelinated nerve fibres in the cochlear nerve of normal hearing individuals and any lower number in cases of sensory neural deafness. There was in general a good correspondence between the counted numbers of the myelinated nerve fibres in the osseous spiral lamina, the spiral ganglion cells and the myelinated nerve fibres in the cochlear nerve in the inner acoustic meatus.

The diameter of the peripheral axons of the type I neurons are about half the diameter of the central axons. The average diameter of the central axons is 2.5 mμ with a narrow distribution in children, but an increasingly larger range of fiber calibers with increasing age (0.5 to 7 μm in the 40 to 50 year age group adults).

Cochlear nerve; Spiral ganglion; Human; Electron microscopy; Quantification; Block surface method

Introduction

Considerable efforts have been made in recent years for quantitative structural analysis of the human cochlea and the cochlear nerve. To promote such efforts, a European Working group for the assessment of normative data in the human cochlea has been established in 1984 within the framework of the Medical Research Program of the European Communities. So far, precise quantitative data have been reported mainly from the cochlea (Pollak et al., 1987; Gleeson and Felix, 1987; Spoendlin and Schrott, 1987; Wright et al., 1987) but very few from the cochlear nerve.

After the original work of Guild (1932) and Rasmussen (1940), some studies were concerned with the gross anatomy and fascicular pattern of the VIIIth and VIIth cranial nerve (Silverstein, 1984; Schefter and Harner, 1986). More recent, finer analysis of the VIIIth nerve have been reported on surgically removed specimens of deaf patients (Ylikosky and Solvainen, 1984) and in some temporal bones from individuals with normal or slightly impaired hearing (Felix et al., 1987). More information is available on the total number of the human spiral ganglion cells, however with considerable differences in the results. The aim of our study was to obtain precise quantitative and qualitative data of the cochlear neurons evaluated at different levels with light and electron microscopy.

Materials and Method

18 human cochleas from 14 patients were evaluated using the block surface technique (Spoendlin and Schrott, 1987). In 14 temporal bones from 10 patients, the VIIIth nerve was carefully analysed. In 5 of these cases both the cochlea and the cochlear nerve were examined. For 16 temporal bones, the pure tone audiogram from within 1 year prior to death was available. 4 adults had a normal pure tone audiogram and 1 adult and 2 children were reported as normal hearing. The others suffered from various degrees of sensory-neural deafness.

Correspondence to: Prof. Dr. H. Spoendlin, H.N.O.-Klinik, University of Innsbruck, Anichstrasse 35, A-6020 Innsbruck, Austria.

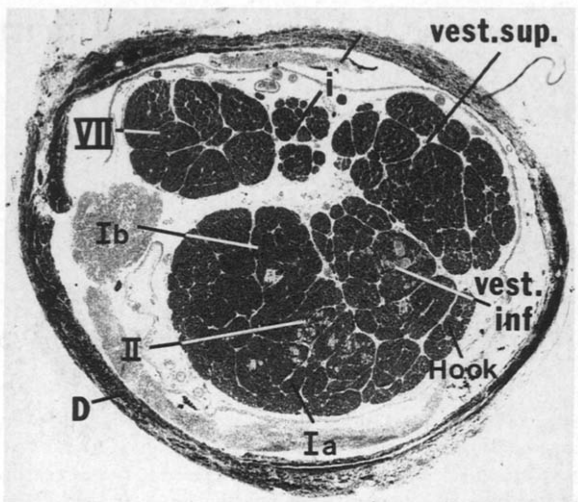

Fig. 1. Transverse section through the intrameatal portion of the VIIIth nerve between level 2 and 3 (see Fig. 4) of a 8 year-old child with the dural sheeth (D) showing the position of the facial nerve (VII), the intermediate nerve (i), the superior devision of the vestibular nerve (vest. sup.), the inferior portion of the vestibular nerve (vest. inf.) and the cochlear nerve with the nerve fibers for the lower end of the cochlea (Hook), for the lower basal turn (Ia), the upper basal turn (Ib) and the second and apical turn (II).

For the evaluation of the cochlea, fixation was done by perilymphatic perfusion with Karnovsky solution as soon as possible, usually within two hours after death, in order to have a sufficiently good quality of fixation (Spoendlin and Schrott, 1987). The fixation quality of the VIIIth nerve was not improved by perilymphatic perfusion. It was rather dependent from the delay between autopsy and immersion fixation in Karnovsky solution. The temporal bones were removed as a bone plug with a cylindric striker saw (Schuknecht, 1968), taking care not to damage the VIIIth nerve. After removal, the specimens underwent a prolonged immersion fixation in Karnovsky solution from one to several days. The VIIIth nerve was then carefully microdissected from the inner acoustic meatus making an effort to keep the nerve as a whole together with the pial sheaths and the blood vessels (Fig. 1). Postfixation with 1% OSO_4 in cacodylate buffer was followed by embedding in spurr epoxy resin. Semithin and ultrathin sections were obtained for light- and electronmicroscopy.

In the block surface preparations of the cochleas and the preparation of the VIIIth nerve, it was possible to assess quantitatively the cochlear neurons at a light- and electronmicroscopic level within the osseous spiral lamina, the spiral ganglion and the cochlear nerve. To study the segmental innervation densities, the cochleas were divided into 10 equal segments, in which the sensory cells were counted in surface microscopy, the nerve fibers of the osseous spiral lamina in tangential sections and the spiral ganglion cells in radial or tangential serial sections (Fig. 2) (Spoendlin and Schrott, 1988).

For the quantitative assessment of the spiral ganglion cells, either the nucleus or the nucleolus can be counted in serial sections. Although the results of the two counting methods did not differ very much in our hands, counting the nucleoli appears to be easier and more accurate, provided each nucleus has always one and only one nucleolus. In 30 μm thick sections we examined 500 spiral ganglion cells, of which the entire nucleus was contained within the section and we found in no occasion a missing nucleolus or more than one nucleolus per ganglion cell. The possibility to double count nucleoli which are divided by the section, is small, because of their small size. In unstained 10 μm thick light microscopic sections, the nucleolus is only clearly seen in its full size when nearly half of it is included in the section (Fig. 3).

By counting the nearly spherical nucleoli in every section of 10 μm thick serial sections, the theoretical correction factor as calculated with the formula of Floderus (1944) or Haug (1967) becomes very small, possibly negligible.

In this formula

$$N = N' \frac{t}{t + D - 2h}$$

N = corrected cell count, N' = actual cell count, t = thickness of the section, D = diameter of the object counted, h = height of the unvisible segments of the counted object.

If the section thickness (t) is 10 μm, the diameter of the nucleoli (D) 2.5 μm and the thickness of the segments of the nucleoli which would not show up clearly in full size in the section (h) is 1 μm, the correction factor will be: $10/(10 + 2.5 - 2) = 0.952$. Because however the decision whether a nucleolus is clearly seen in its full size, is essen-

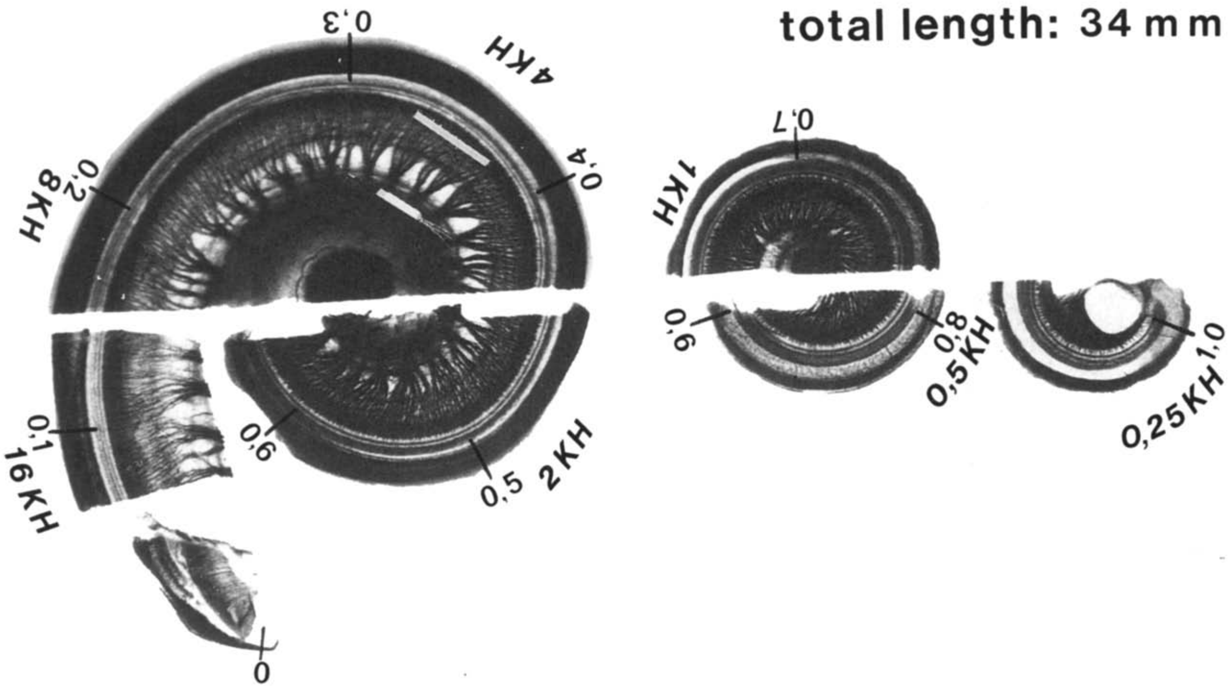

Fig. 2. Block surface preparation of the cochlea of a 47 year-old man with a normal pure tone audiogram. The cochlea is divided in 10 equal segments, each one 1/10 of the total cochlear length (0.1–1.0). The corresponding frequency ranges are indicated in kHz. The white bars in the lower basal turn in the area of 4 kH show the corresponding areas of the osseous spiral lamina and the spiral ganglion in which the numbers of nerve fibers and the numbers of spiral ganglion cells can be correlated. The correction factor for the spiral ganglion cells for this particular area would be 0.6.

Fig. 3. Some spiral ganglion cells in interference contrast microscopy. Type I ganglion cells (I) are about double the size of the type II ganglion cells (II). In two of the type I ganglion cells, the nucleolus is clearly seen in its full size (N). In all the other cells it is either not seen at all or only incomplete.

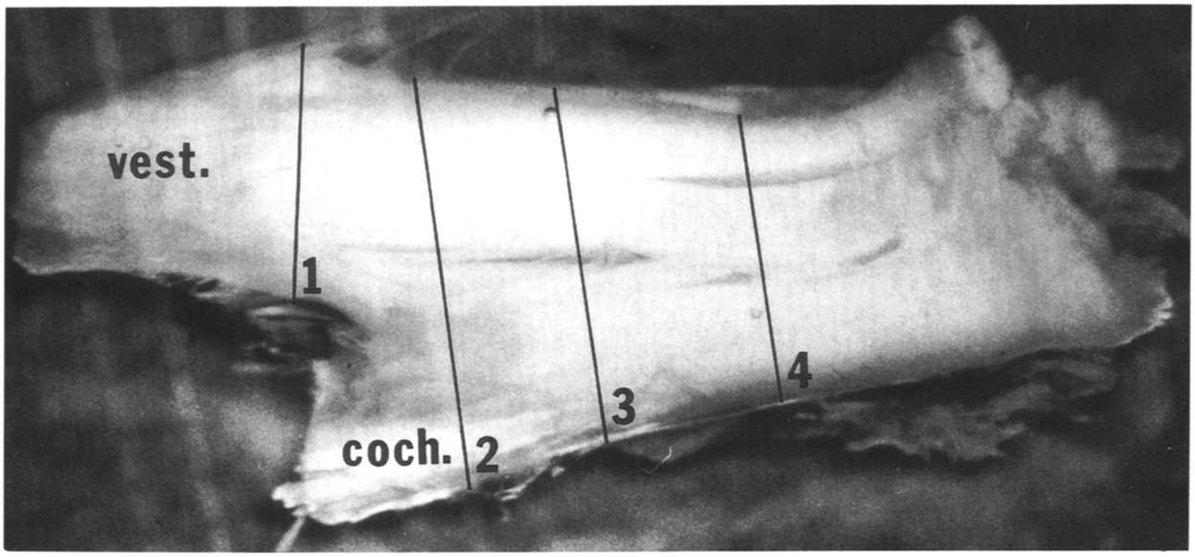

Fig. 4. VIIIth nerve of an adult normal hearing man, carefully dissected from the inner acoustic meatus, seen from posterior with the vestibular nerve (vest.) and the cochlear nerve (coch.). 1–4 show different levels at which the nerve was analysed in transverse sections.

tially a subjective estimate, the evaluated numbers might still differ to a small extent from the real numbers.

The cochlear nerve was evaluated in transverse sections at different levels (Fig. 4). The fibers were counted on light microscopic fotographic reconstructions of transverse nerve sections enlarged by a factor of 300 or with a computer aided morphometric program (Vids IV AI-Tectron, F.R.G.). For caliber measurements and structural details, light microscopic enlargements of 800 times or electron micrographs of selected areas of the nerve were used.

Results

Osseous spiral lamina

As seen in 5 cochleas of normal hearing persons the number of myelinated axons in the lamina per millimeter length of the cochlear duct, which indicates the innervation density of the organ of Corti, varies considerably from base to apex. It is minimal with 300 fibers per millimeter at the basal end of the cochlea, reaches a maximum in the lower second turn (in the 0.5–0.7 region) with 1.400 fibers per millimeter and decreases again towards the cochlear apex to about 400 fibers per

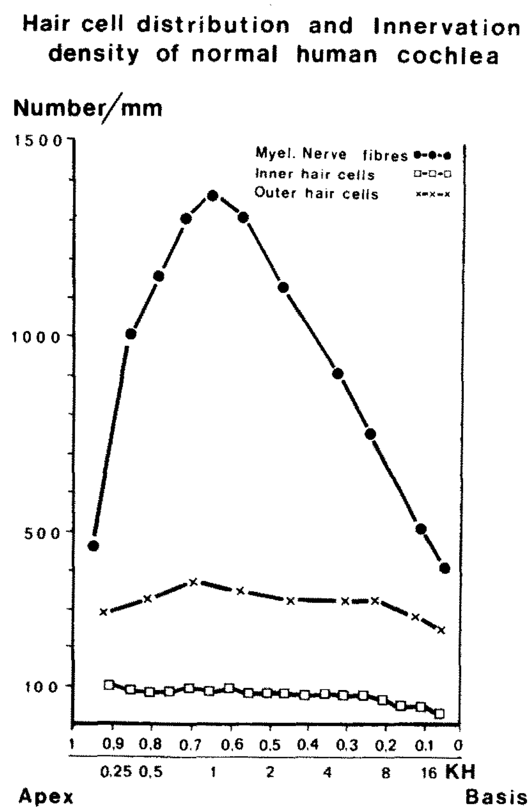

Fig. 5. Hair cell distribution and innervation density of a normal human cochlea.

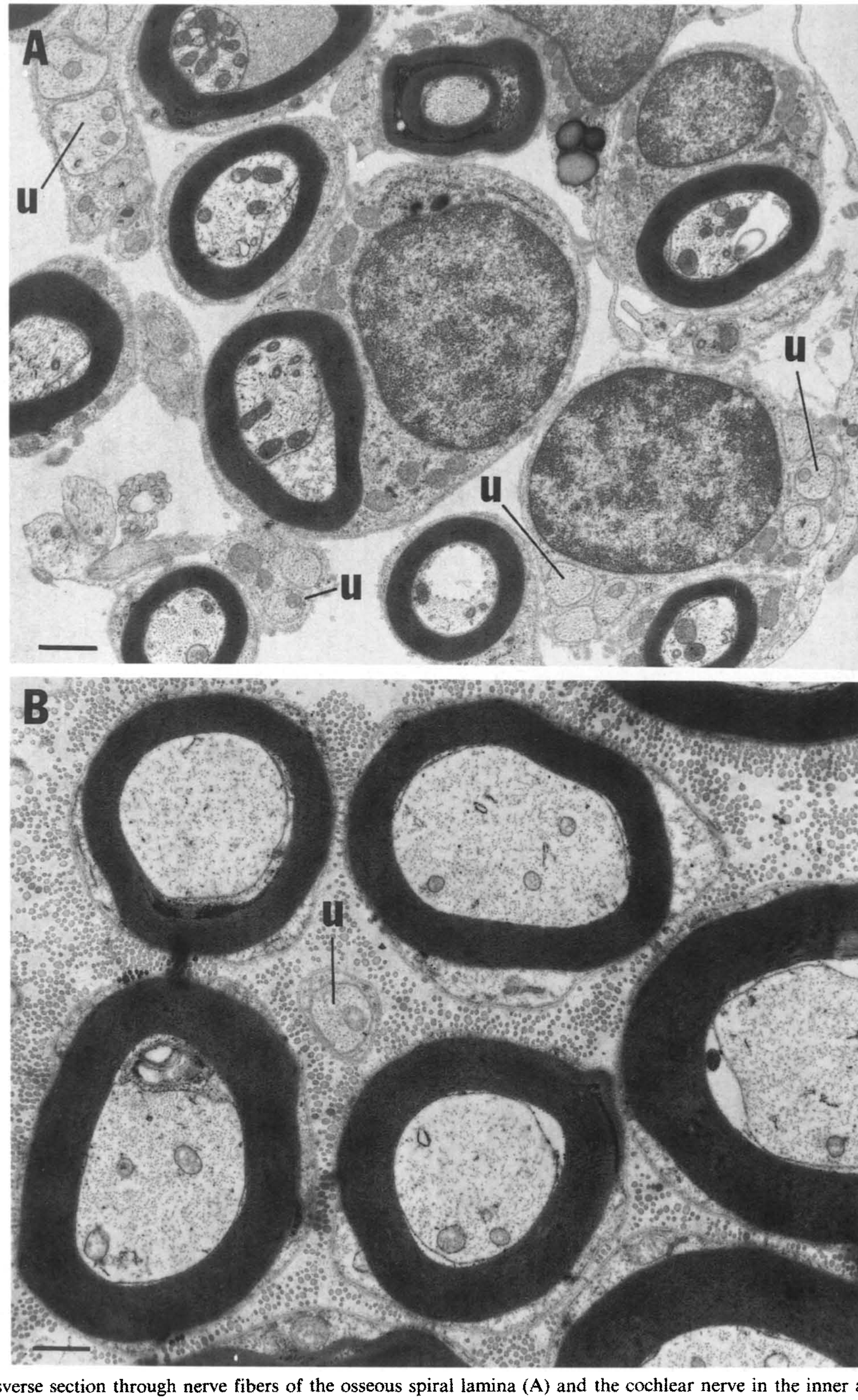

Fig. 6. Transverse section through nerve fibers of the osseous spiral lamina (A) and the cochlear nerve in the inner acoustic meatus (B) of a normal hearing 8 year-old child. Both figures are shown with the same magnification. The pronounced difference in size between the myelinated nerve in the osseous spiral lamina (A) and the fibers in the cochlear nerve (B) is clearly evident. In the osseous spiral lamina there are many more unmyelinated nerve fibers (u) than in the cochlear nerve. Scale bars = 1 μm.

millimeter (Fig. 5). This corresponds to an average of 15 nerve fibers per inner hair cell in the lower second turn and 3–4 nerve fibers per inner hair cell at the base and apex.

In general, the majority of the fibers are myelinated with a diameter beetween 1 and 2 μm and a fairly narrow unimodal distribution (Fig. 6A). There are no significant differences in the size of the nerve fibers of different cochlear turns. The lengths of the peripheral axons between the organ of Corti and Spiral ganglion is with about 1.4 mm in the lower basal and 1.1 mm in the upper basal and second turn in the same order of magnitude in all cochlear turns. Compared to the central axons, the myelin sheath is relatively thin with 20 to 23 myeline lamellae (0.25 μm).

The percentage of the unmyelinated fibers varies from 10% to over 50% in certain regions (Fig. 6A). In the basal turn there are more unmyelinated fibers than in the second and third turn. Their diameters vary from 0.2 to 1.5 μm, the majority being between 0.5 and 1 μm. Usually, two or more unmyelinated nerve fibers are surrounded by the extensions of one single Schwann cell.

Spiral ganglion

Two types of ganglion cells, the type I and type II cells can easily be distinguished in light microscopic preparations. Both types are mostly unmyelinated. The type I cells represent about 90% of the ganglion cell population. They have a diameter of 25–30 μm, their nucleus is spherical with a very constant diameter between 11–12 μm, a loose unevenly distributed chromatin and a very pronounced nearly spherical nucleolus with an average diameter of 2.5 μm (Fig. 7).

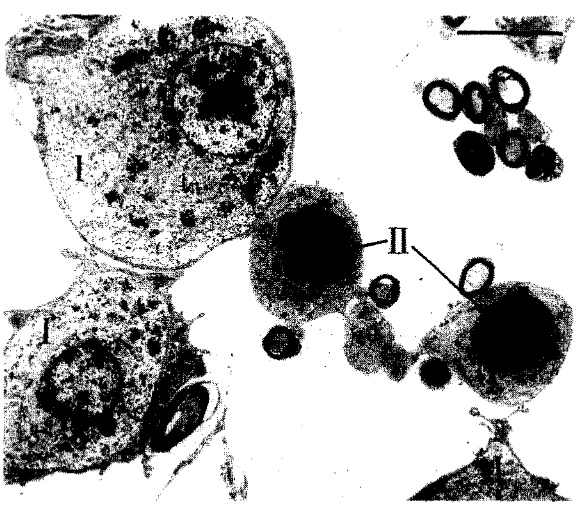

Fig. 7. Spiral ganglion cells of type I (I) and type II (II) of a normal hearing adult person showing the pronounced difference between the two types of ganglion cells. Scale bar = 10 μm.

The type II cells represent about 10% of all spiral ganglion cells in normal individuals. They are mostly situated in the peripheral portion of the spiral ganglion and are much smaller with an average diameter of 15 μm. They have a spherical nucleus with a diameter of 7.5 μm, a very homogenous chromatine and a very small, not very pro-

COCHLEAR NEURONS

	DIST.FROM BASE / LENGTH	LAMINA NF / MM	SP.GGL.CELLS/MM TYPE I	TYPE II	DIFF. TYPE I / NF
NORMAL 8 Y.	0.3	826	860	91 (11%)	+ 4%
	0.4	1375	1420	93 (7%)	+ 3%
	0.6	1403	1520	109 (7%)	+ 8%
40 Y.	0.1	550	520		- 5%
	0.3	1207	1144	77 (7%)	- 5%
	0.4	1290	1200		- 7%
	0.5	1404	1387	95 (7%)	- 1%
49 Y.	0.3	650	665	100 (15%)	+ 2%
	0.5	1156	1183	75 (6%)	+ 2%
NEURAL PRESBY 62 Y.	0.2	236	227	72 (31%)	- 4%
	0.4	744	764	80 (10%)	+ 3%
	0.6	930	855	105 (12%)	- 8%
SENS. PRESBY 68 Y.	0.2	467	443	73 (16%)	- 5%
	0.3.	681	664		- 2%
	0.4	841	864		+ 3%
	0.6	1372	1416	106 (8%)	+ 2%

Fig. 8. Correlation of the counted numbers of myelinated nerve fibers in the osseous spiral lamina and the spiral ganglion cells corrected with the correction factor in different cochlear segments (see Fig. 2). The difference between the counted figures of type I ganglion cells and nerve fibers never exceeded 8%. The percentage of type II ganglion cells increases in cases of neural and sensory presbyacusis. There is good correspondence between the number of nerve fibres in the lamina and the spiral ganglion cells not only in normal hearing individuals but also in the cases of neural and sensory presbyacusis.

nounced nucleolus of 0.75 μm diameter. Further distinguishing ultrastructural characteristics are the great amount of filaments and relatively few ribosomes in the cytoplasm of type II cells (Fig. 7), in contrast to the cytoplasm of type I cells, which contains more ribosomes, and very few filaments (Spoendlin and Schrott, 1988). In cases of sensory neural deafness with primary or secondary neuronal degeneration, the total number of the type II cells remains more or less unchanged and their relative number increases (Fig. 8).

The segmental distribution of the spiral ganglion cells was determined in 5 cochleas of 4 adults and 1 child with normal hearing and in 2 cochleas of adults with sensory-neural deafness. In normal hearing persons, there are from 1000–1200 ganglion cells per millimeter cochlear segment in Rosenthal's canal in the lower basal turn and 2000–2800 ganglion cells per millimeter in the mid cochlear region. In cases of sensory-neural deafness much fewer ganglion cells are found. Using an appropriate correction factor, the numbers can be correlated with the numbers of nerve fibers in the osseous spiral lamina of the corresponding cochlear segment (Figs. 2 and 8). There is, in general, a good correlation between the number of type I ganglion cells and the myelinated nerve fibers in the osseous spiral lamina. Such correlations are however only possible in the basal and lower second turn. In the upper turns, the spiral ganglion consists of one central group of cells which can not be accurately related to the segments of the osseous spiral lamina or the organ of Corti (Spoendlin and Schrott, 1988).

For the evaluation of the total number of ganglion cells, horizontal serial sections were used. In the cochlea of a 49 year old man with a normal

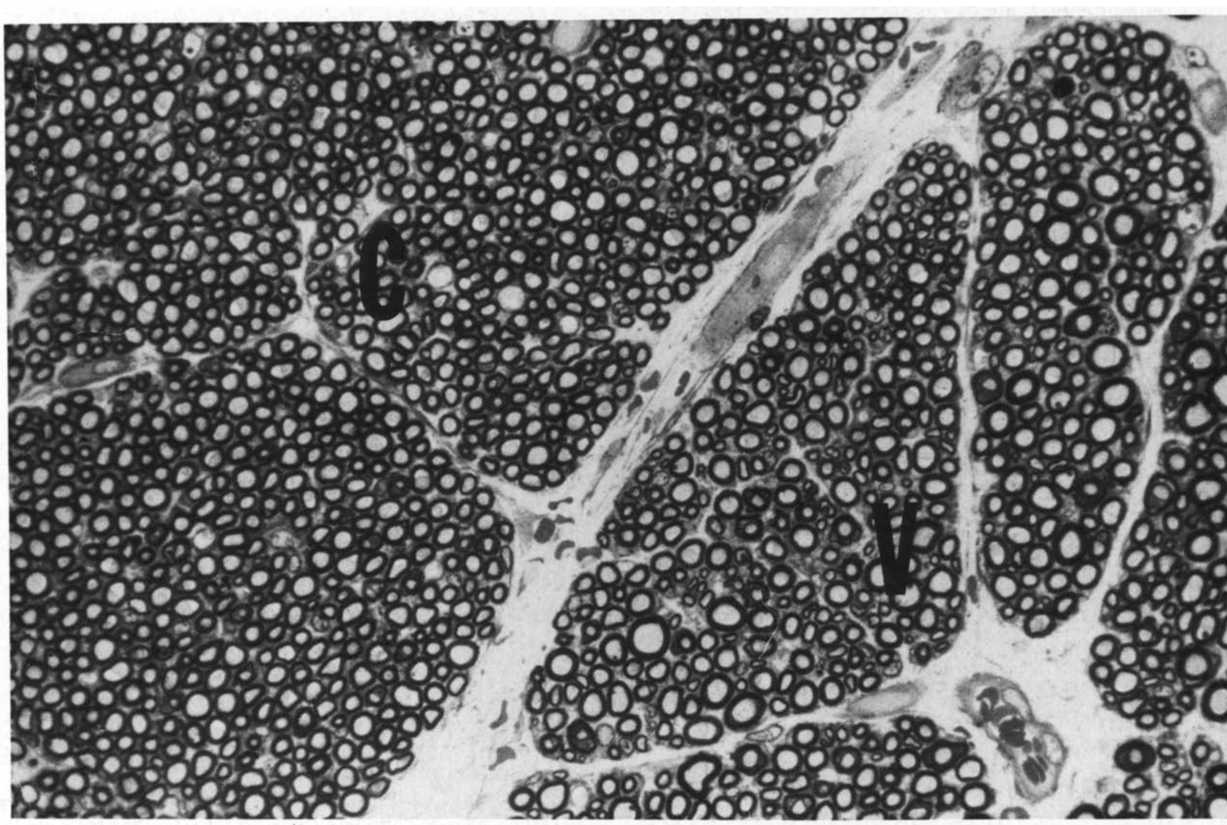

Fig. 9. Light microscopic picture of an adjacent portion of the cochlear nerve (C) and vestibular nerve (V) of a 8 year-old normal hearing child. The average size of nerve fibers in the cochlear nerve is smaller than in the vestibular nerve and the range of caliber size is much larger in the vestibular nerve (see also Fig. 10).

pure tone audiogram we found a total number of 30,000 type I ganglion cells, which correspond relatively well with the counted 31,000 myelinated nerve fibers in the cochlear nerve.

Cochlear nerve

There is frequently no clear anatomical separation between cochlear and vestibular nerve, and their exact delimitation can be very difficult, especially at the more central levels (Fig. 1). However, the average diameter of the myelinated nerve fibers is significantly smaller in the cochlear nerve than in the vestibular nerve and this difference in diameter can help to differentiate between cochlear and vestibular neurons (Figs. 9 and 10) The topographic arrangement of the fibers of the cochlear nerve mimics the spiral of the cochlea. The fibers for the lower basal turn are situated in the tale portion of the nerve, the ones for the apex in the center of the main portion (Fig. 1).

The total number of myelinated nerve fibers in the cochlear nerve of normal hearing persons varies in our material from 36,000 to 41,000 in 3 cochlear nerves from 2 children, 32,000 to 37,000 in 4 cochlear nerves from 3 adults and any lower figure in cases of sensory neural deafness. There are generally no significant differences between the

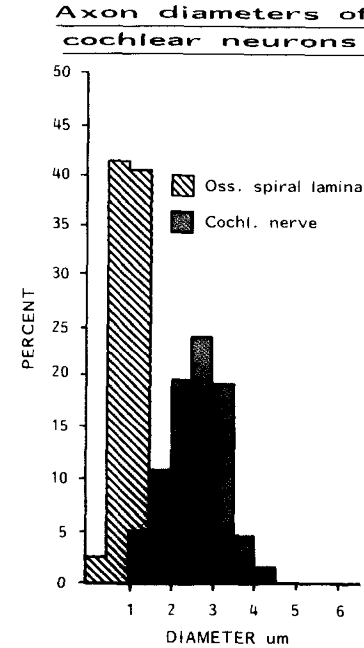

Fig. 11. Histogram of the fiber caliber of myelinated nerve fibers in the osseous spiral lamina and the intrameatal cochlear nerve.

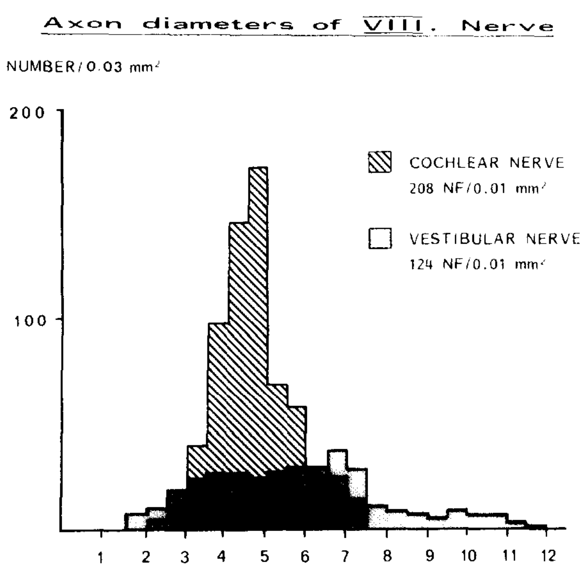

Fig. 10. Histogram of the caliber of cochlear and vestibular nerve fibers of a 8 year-old normal hearing child.

nerve fiber counts at different levels of the cochlear nerve, thus excluding substantial fiber ramification. In one case where the spiral ganglion as well as the cochlear nerve were evaluated, we found a good correlation between the total number of spiral ganglion cells and the number of myelinated nerve fibers in the cochlear nerve.

Accurate structural evaluation is only possible in well fixed material, where the myelin sheath has a regular annular appearence and the axons show only minor shrinkage. In less well fixed material, where the annular shape of the myelin sheaths becomes irregular and the axons show considerable shrinkage, the fibers tend to be larger. Measurements in well fixed nerves done by two different persons with different methods showed a maximum variation of $\pm 10\%$.

The average diameter of the myelinated nerve fibers in the cochlear nerve is about double the value of the myelinated fibers in the osseous spiral lamina (Figs. 6B and 11). There is an unimodal distribution of the fiber diameters which appear to be much more uniform in children (Fig. 12) than in adults where we found a much wider range with

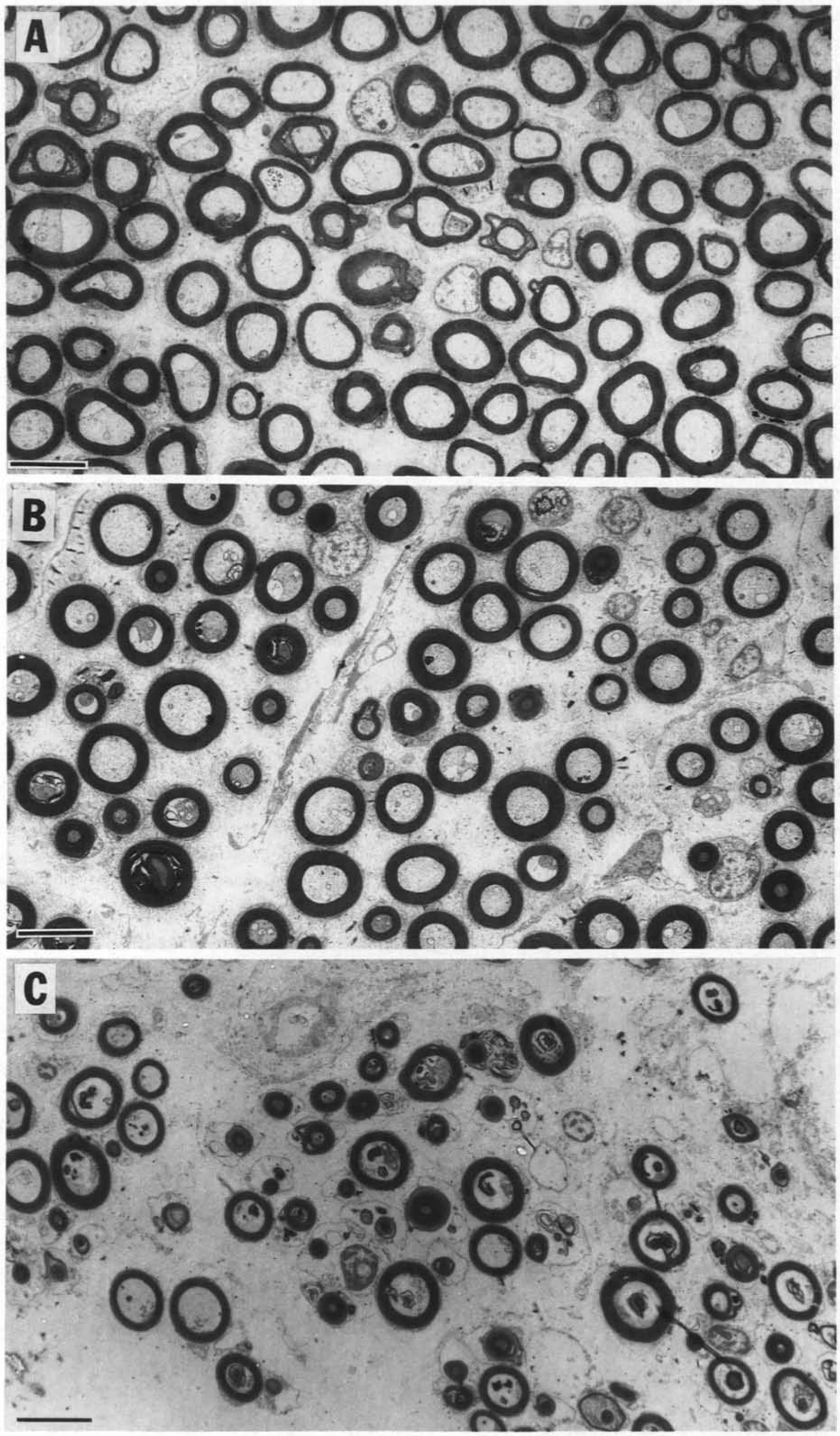

Fig. 12. Electron microscopic pictures of myelinated cochlear nerve fibers of the intrameatal segment of the cochlear nerve in a 8 year-old normal hearing child (A), a 48 year-old normal hearing adult (B) and an adult with sensory neural deafness (C). (see also Figs. 13 and 15). Scale bars = 5 μm.

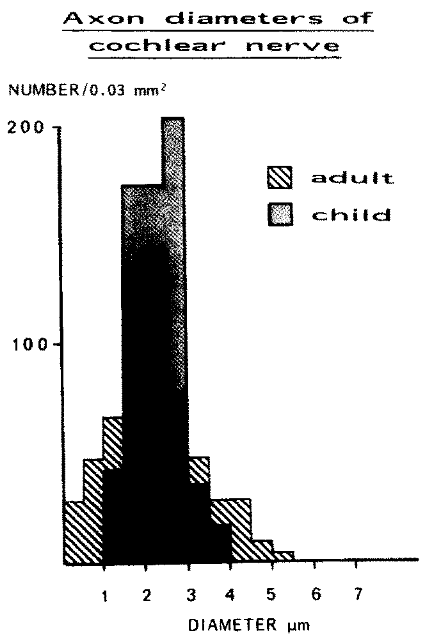

Fig. 13. Comparison of the caliber of myelinated cochlear nerve fibers in a normal hearing child and a normal hearing adult.

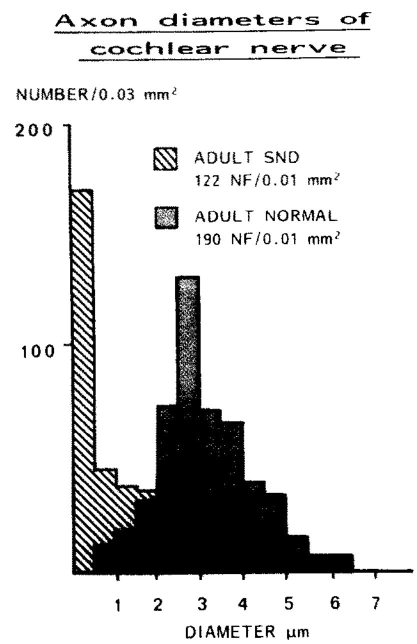

Fig. 15. Histogram of the caliber of myelinated nerve fibers in a normal hearing adult and an adult with sensory neural deafness.

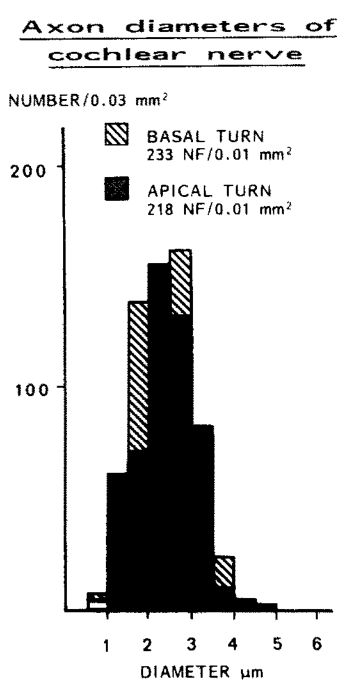

Fig. 14. Histogram of the caliber of myelinated cochlear nerve fibers of the basal and apical turn of a normal hearing 8 year-old child.

is a clear peak between 2 and 3 μm (Fig. 13). In general, we measured the inner diameters (axon without myelin sheath). In cases where we measured the inner and outer diameters (including the myelin sheath) we found no significant difference in the distribution pattern. There is also no significant difference between the fibers of the cochlear apex and the cochlear base, except the fact that there are greater irregularities and reduced fiber densities at the basal end (Fig. 14). In patients with sensory neural deafness, the variation of the fiber calibers is more pronounced and the proportion of small fibers is increasingly larger. In severe cases, a considerable number of nerve fibers consists only of myelin with no visible or extremely small pathological axons (Figs. 12c and 15).

With few exceptions, the myelin sheath consists of 60 to 83 myelin lamellae which amounts to 0.7 to 1 μm. Unmyelinated nerve fibers can not be detected in light microscopic sections. At an ultrastructural level we found only 2–4% unmyelinated nerve fibers within the cochlear nerve (Fig. 6B).

Discussion

The exact quantitative and qualitative evaluation of the human cochlear nerve at the level of the osseous spiral lamina, the spiral ganglion and the cochlear nerve is of considerable interest in respect to electrocochleography and BAEP-recordings, direct recordings from the cochlear nerve (Møller and Jannetta, 1981; Møller et al., 1988) and electrical stimulation of the cochlear nerve.

In contrast to a frequently expressed opinion, we found a relatively good correlation between the number of axons in the osseous spiral lamina and the spiral ganglion cells in corresponding cochlear segments. The good correspondence of the number of myelinated nerve fibres in the osseous spiral lamina with the number of spiral ganglion cells in our 2 cases with pronounced sensory-neural deafness (Fig. 8) suggests that in the course of retrograde degeneration the peripheral axons of the majority of type I neurons disappear only when the pericaryons in the spiral ganglion degenerate. Also in experimentally induced retrograde degeneration in animal studies we found in general good correspondence between the numbers of myelinated nerve fibres in the osseous spiral lamina and the spiral ganglion cells, although degeneration started some weeks earlier in the nerve fibres of the lamina (Spoendlin, 1975, 1984).

As in other mammals, also in the human, type I and type II neurons in the spiral ganglion can clearly be distinguished. In analogy to the situation in all so far studied mammals, we can assume that the type II neurons which represent only 10 % of all cochlear neurons are associated with the outer hair cells. In the human however, their peripheral axons have a much longer spiral extension at the level of the outer spiral fibers and more peripheral ramifications than in other mammals (Spoendlin and Schrott, 1988). Most probably not only their peripheral axons, but also their central axons are unmyelinated. However, within the cochlear nerve, unmyelinated axons are not only very small but also very few (Spoendlin, 1985), which illustrates the poor connection of the type II neurons to the central nervous system. On the basis of studies with combined recordings and neuron labeling (Liberman, 1982), we must assume that electrophysiological recordings catch almost exclusively the activity of the myelinated type I neurons in the cochlear nerve. In cases of severe sensory neural deafness, it appears that the type II neurons persists in greater numbers than the type I neurons. They seem to be more resistent to retrograde degeneration (Spoendlin and Schrott, 1988).

Our figures of the total number of myelinated nerve fibers in the cochlear nerve in normal hearing persons are slightly higher with smaller differences between the maximum and minimum values than the figures reported in the literature by Rasmussen (1940) and by Felix et al. (1987). This might be due to the relatively poor tissue preservation in Rasmussen's material, the method of counting and the difficulties to critically delineate the cochlear and vestibular nerve.

Numerous reports exist on the spiral ganglion cell counts in serial sections of celloidin embedded material (Guild et al., 1931; Guild, 1932; Otte et al., 1978; Hinojosa et al., 1985; Pauler et al., 1986; Pollak et al., 1987; Nadol, 1988). The average reported figures for normal hearing persons vary from 17,380 to 39,114 in all age groups and from 21,142 (Felix et al., 1987) to 39,114 (Guild et al., 1931) in adults. The values from Hinojosa, who also counted the nucleoli, correspond best with our results. Correlations between ganglion cell counts and nerve fiber counts in the cochlear nerve were only done by Felix et al. (1987). Surprisingly, they found up to 24% less spiral ganglion cells than fibers in the cochlear nerve. In one case of a normal hearing person in which we carefully evaluated the spiral ganglion and the cochlear nerve, we found only a difference of 6% between type I ganglion cells in the spiral ganglion and the myelinated nerve fibers in the cochlear nerve. This small difference can be accounted for by counting errors and myelinated efferent fibers.

Segmental evaluation of ganglion cells was carried out by Spoendlin and Schrott (1988) and Nadol (1988). Nadol, by calculating the number of ganglion cells per 0.001 mm^3 of spiral ganglion, found the greatest density of spiral ganglion cells in the mid region of the cochlea. In our study we correlated the number of ganglion cells to the number of inner hair cells in corresponding cochlear segments and found the greatest innervation density in the 20 to 24 mm region (lower second turn) (Figs. 5 and 8). However, since the number of spiral ganglion cells and the number of myelinated nerve fibers in the osseous spiral lamina of a given cochlear segment do not differ significantly, it is easier and more accurate to determine the innervation density by counting the number of myelinated nerve fibers in the osseous spiral lamina rather than counting the spiral ganglion cells in that segment. Among the peripheral axons there are up to 50% unmyelinated axons in contrast to 2–4% among the central axons of the cochlear nerve.

Although direct demonstration is not possible in man, we must assume in analogy to the situation in mammals, that the myelinated nerve fibers represent the afferent neurons associated with the inner hair cells and the efferents to the outer hair cells, whereas the unmyelinated fibers represent the afferent neurons to the outer hair cells and the efferent neurons to the inner hair cells as wells as some adrenergic autonomic nerve fibers (Spoendlin, 1971, 1985).

As already noticed in the cat, the caliber of the peripheral axons of the myelinated type I cochlear neurons is definitely smaller than the caliber of the central axons (Kiang et al., 1982). In the human, the average diameter and the thickness of the myeline sheath of the central axon is about double that of the peripheral axon. This caliber difference might be in relation to the length of the axons. The average lenght of the peripheral axon in the human is 1.2 mm, whereas the total length of the central axon varies between 30 and 40 mm, the intrameatal portion being 8 to 10 mm.

The density of nerve fibers varies considerably in the cochlear nerve of different normal hearing cases from 140 to 240 nerve fibers per 0.01 mm^2. In children, the density is higher than in adults. The range of fiber diameters is larger in cases with lower fiber density. In all our cases, the average and the range of diameters is much larger in the vestibular nerve as compared to the cochlear nerve.

The mean axon diameter of the intrameatal portion of the cochlear nerve of 5 normal subjects, was found to be fairly constant between 2.7 and 3.1 μm, with the exception of more smaller fibres at the lower cochlear end, in the hook area, where pathological alterations can occur early in life without effect on audiometric pure tone threshold in the usually measured frequency range. The much larger values of Rasmussen can not be compared with our results, because they reflect the pronounced tissue alteration by the formalin fixation used at his time.

The axon diameter has a direct influence on the speed of action potential propagation. The larger the fiber, the faster the propagation. For axons with a diameter of 15 μ, conduction velocities of 100 m/s and for axons with a diameter of 3 μm, only velocities of 10–20 m/s have been measured (Gasser and Grundfest, 1939; Dudel, 1987).

As time of transmission plays a very important role in the cochlea and the cochlear neurons,where synchronisation of activities and periodicity pitch sensation are important mechanisms, the caliber of the cochlear nerve fibers is certainly an important factor in the system. We must assume that great differences in fiber calibers goes along with considerable differences of conduction velocity.

The fiber spectrum of the cochlear nerve is larger in the human than in most animals (Arnesen and Kjelsberg-Osen, 1978; Friede, 1984; Spoendlin and Schrott, 1988; Anniko and Arnesen, 1988) ranging from 1 to 6 μm in children and from 0.5 to 7 μm in adults. There is a clear difference in the distribution of fiber calibers between normal hearing children and adults. In children the great majority of fiber diameters are within a narrow range whereas in adults there is a wider range and a much larger proportion of very small or very large fibers, although the mean diameter is equal for both. As a functional consequence, we would expect much greater differences of conduction velocity between individual fibers in adults than in children. This might not affect so much the pure tone threshold hearing but rather more complex auditory functions such as speech discrimination. This however remains an assumption for the moment, because we have only very few temporal bones of patients who had speech audiometry done. In cases of sensory neural deafness with poor speech discrimination, not only the total number of fibers decreases, but also the percentage of very small fibers with diameters below 1 μm increases enormously (Figs. 12 and 15).

Significant caliber differences between the nerve fibers derived from the basal region, and nerve fibers derived from the apical region of the cochlea, have been reported in the guinea pig (Friede, 1984), and in the mouse (Anniko and Arnesen, 1988). Friede found the apical fibers in the guinea pig half as thick as the basal fibers, whereas Anniko and Arnesen describe the apical fibers in the mouse somewhat larger than in the basal fibers. In the human, we were not able to find significant fiber spectrum differences between the apical and basal fibers. An explanation of this difference in findings is lacking. It might however be due to the fact that in the guinea pig the apical nerve fibers are much longer than the basal fibers, whereas in the human, there are not many differences in the length of the fibers. A direct correlation of the nerve caliber to the tonotopical organization of the cochlea as assumed by Friede (1984), is however very unlikely.

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